

# Week 4: Customer Order and Purchase Analysis System

## 1. Overview

The Organic Grocery Store uses a database to manage customer orders and product purchases. It helps keep track of sales, customer activities, and product demand. The system makes it easier to review purchase details and analyse customer spending.

## 2. Database Tables

Customer Details: Stores information about customers, including their ID, name, contact number, email, and address.

Order Records: Maintains details of each purchase, such as order number, customer ID, purchase date, total cost, and delivery status.

Order Information: Records the products included in each order, along with their quantity and price.

Product Details: Contains information about the organic items sold in the store, including product names, prices, and stock availability.

## 3. Customer Order History Report

This report provides a summary of customer purchases. It displays customer names, order numbers, dates, amounts, and order status.

```
SQL> SELECT c.first_name, o.order_id, o.order_date, o.total_amount, o.order_status FROM Customer c JOIN Orders o ON c.customer_id = o.customer_id;
```

| FIRST_NAME   |     | ORDER_ID     | ORDER_DAT     |
|--------------|-----|--------------|---------------|
| TOTAL_AMOUNT |     | ORDER_STATUS |               |
| Arun         | 500 | Delivered    | 101 01-SEP-26 |
| Priya        | 750 | Pending      | 102 02-SEP-26 |
| Arun         | 300 | Shipped      | 103 03-SEP-26 |
| FIRST_NAME   |     | ORDER_ID     | ORDER_DAT     |
| TOTAL_AMOUNT |     | ORDER_STATUS |               |
| Kavin        | 900 | Delivered    | 104 04-SEP-26 |
| Priya        | 400 | Pending      | 105 05-SEP-26 |

#### 4. Product-wise Order Report

This report shows how many times each organic product was ordered and the total quantity sold.

```
SQL> SELECT p.product_name, COUNT(od.product_id) AS times_ordered, SUM(od.quantity) AS total_quantity FROM Product p JOIN Order_Details od ON p.product_id = od.product_id GROUP BY p.product_name;
```

PRODUCT\_NAME

-----

TIMES\_ORDERED TOTAL\_QUANTITY

-----

Organic Carrot

1

5

Organic Banana

1

2

Brown Rice

1

2

PRODUCT\_NAME

-----

TIMES\_ORDERED TOTAL\_QUANTITY

-----

Organic Apple

2

3

Green Tea

1

3

#### 5. Customer Purchase Analysis

##### A. Customers with Maximum Orders

This query finds the customer who placed the highest number of orders.

```
SQL> SELECT customer_id, COUNT(*) AS total_orders FROM Orders GROUP BY customer_id HAVING COUNT(*) = ( SELECT MAX(order_count) FROM ( SELECT COUNT(*) AS order_count FROM Orders GROUP BY customer_id ) );
```

CUSTOMER\_ID TOTAL\_ORDERS

-----

1

2

2

2

##### B. Total Spending by Each Customer

This query calculates the total amount spent by each customer.

```
SQL> SELECT c.first_name, SUM(o.total_amount) AS total_spending FROM Customer c JOIN Orders o ON c.customer_id = o.customer_id GROUP BY c.first_name;
```

| FIRST_NAME | TOTAL_SPENDING |
|------------|----------------|
| Kavin      | 900            |
| Priya      | 1150           |
| Arun       | 800            |

### C. Average Order Value

This query calculates the average order amount for each customer.

```
SQL> SELECT c.first_name, AVG(o.total_amount) AS average_order_value FROM Customer c JOIN Orders o ON c.customer_id = o.customer_id GROUP BY c.first_name;
```

| FIRST_NAME | AVERAGE_ORDER_VALUE |
|------------|---------------------|
| Kavin      | 900                 |
| Priya      | 575                 |
| Arun       | 400                 |